

The Title is Not Dog Treats

By

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The Title is Not Dog Treats

Chair: Perry Alexander

Date approved: _____ MM DD, YYYY

Abstract

Formal verification is a useful tool in guaranteeing the security and behavior of programs. This is increasingly true as the amount of LLM generated code continues to rise. However even simple projects can be labor intensive to verify, thus verification see's little use in practice. Existing methods have either used program synthesis algorithms or LLM's to automate verification. We continue this work with NotDogTreats an LLM based proof synthesizer for Verus which iteratively attempts to constructs proofs utilizing prompts containing relevant information about the problem.

Acknowledgements

This is where you will insert your acknowledgements. Usually in this section, students will mention their thesis/dissertation chair and committee, any research associates, their friends, and their family members. But who you thank is ultimately up to you.

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Chapter 1

Introduction

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

1.1 Overview

Type out the content of your subsection here. What follows is an example APA 7th citation (Metz, 1978). Note also Metz, Einstein, et al. (1978) but Metz, Hamlin, et al. (1978). You can adjust citational information by editing the allcites.bib file, included in this package.

1.2 Background?

Discuss formal methods and proof synthesis. What someone might need to know to understand this paper, How did the field develop to where we are today, and the significance of the field.

Chapter 2

Literature Review

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

2.1 Name of Subsection

Type out the content of your subsection here (Einstein, 1905). Delete or duplicate as necessary.

Note that you can also refer to figures in the text like this: see figure 2.1.

Figure 2.1: Insert figure caption here



2.1.1 Name of Third-Level Subsection

Here is an example of a third-level subsection. Delete or duplicate as necessary.

2.1.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 3

Methodology

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

3.1 Name of Subsection

Type out the content of your subsection here. Here's another citation example (Littell et al., 1996; Metz, 1978; Wellek, 2003). And here is an example equation:

$$D = 2.91L^{0.71} \tag{3.1}$$

3.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

3.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

3.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 4

Findings

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

4.1 Name of Subsection

Type out the content of your subsection here. Here's another citation (Einstein, 1905; Thomas, 1998). What follows is an example table:

Table 4.1: Insert table caption here

Set	Test 1	Test 2	Test 3	Test 4
1	50	34	837	970
2	47	877	34	230
3	5	31	25	415
4	35	27	144	2356
5	45	300	556	389

And just to be safe, here is another example of a table:

Table 4.2: Insert table caption here

Experiment	Batch A	Batch B	Batch C	Batch D	Batch E
5	7	5	3	7	2
8	9	8	2	1	9
3	2	1	9	6	6
2	2	8	1	4	9

You can also refer to tables in the text like this: see tables 4.1 and 4.2.

4.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

4.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

4.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 5

Conclusion

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

5.1 Name of Subsection

Type out the content of your subsection here. Delete or duplicate as necessary.

5.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

5.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

5.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

References

- Einstein, A. (1905). Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies]. *Annalen der Physik*, 322(10), 891–921. <https://doi.org/10.1002/andp.19053221004>
- Littell, R. C., Milliken, G. A., Stroup, W. W., & Wolfinger, R. D. (1996). *SAS system for mixed models*. SAS Institute, Inc.
- Metz, C. E. (1978). Basic principles of ROC analysis. *Seminars in Nuclear Medicine*, 8, 283–298.
- Metz, C. E., Einstein, A., Thomas, P. A., Benedict, A. B., & Rice, C. (1978). Principles of potentially redundant citations. *Bibliographic Studies*, 34, 56–78.
- Metz, C. E., Hamlin, H., Thomas, P. A., Boss, H., Benedict, A. B., & Redfield, C. R. (1978). A meta-analysis of redundant citations. *Bibliographic Studies*, 34, 79–84.
- Thomas, L. (1998). Statistical power analysis software. <http://www.forestry.ubc.ca/conservation/power/>
- Wellek, S. (2003). *Testing statistical hypotheses of equivalence*. Chapman & Hall/CRC Press.

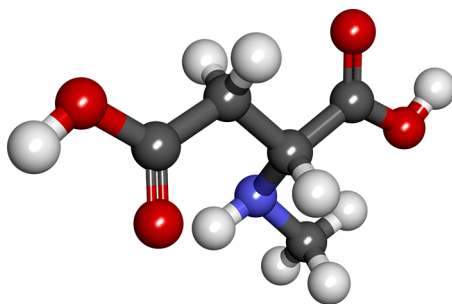
Appendix A

Title of Appendix

A.1 Name of Subsection

This is an example of an appendix. You can put citations here (Metz, 1978), as well as figures:

Figure A.1: Insert figure caption here



A.2 Name of Subsection

Here is another example of an appendix subsection. Delete or duplicate as necessary.

A.2.1 Name of Third-Level Subsection

Here is another example of a third-level appendix subsection. Delete or duplicate as necessary.

A.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level appendix subsection. Delete or duplicate as necessary.

Appendix B

Title of Appendix

B.1 Name of Subsection

This is an example of an appendix. Just like chapters, you can also include tables in appendices:

Table B.1: Insert table caption here

Experiment	Batch A	Batch B	Batch C	Batch D	Batch E
6	9	5	3	7	2
8	3	8	2	1	9
9	7	1	9	6	6
9	4	8	1	4	9

B.2 Name of Subsection

Here is another example of an appendix subsection. Delete or duplicate as necessary.

B.2.1 Name of Third-Level Subsection

Here is another example of a third-level appendix subsection. Delete or duplicate as necessary.

B.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level appendix subsection. Delete or duplicate as necessary.